

HyPiDecoder: Hybrid Pixel Decoder for Efficient Segmentation and Detection

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Abstract

Recently, *Mask2Former* has achieved significant success as a universal image segmentation framework, with its *Multi-Scale Deformable Attention (MSDeformAttn) Pixel Decoder* becoming a widely adopted component in current segmentation models. However, the inefficiency of *MSDeformAttn* has become a performance bottleneck for segmenters. To address this, we propose the *Hyper Pixel Decoder (HyPiDecoder)*, an improved *Pixel Decoder* design that replaces parts of the *MSDeformAttn* layers with convolution-based FPN layers, introducing explicit locality information and significantly boosting inference speed. Experimental results show that *HyPiDecoder* can be applied to both universal segmentation models and unified segmentation and detection models, achieving improvements in both speed and accuracy across object detection, semantic, instance, and panoptic segmentation tasks. The *Mask DINO* model integrated with *HyPiDecoder* achieves a new SOTA of 58.8 PQ on COCO panoptic segmentation with SwinL-scale backbone and no extra training data, with a 127% increase in inference speed compared to the original model. Code is available at <https://github.com/HyPiDecoder/HybridPixelDecoder>.

1. Introduction

Segmentation and object detection are both important tasks in the field of computer vision. Both tasks aim to localize objects of interest in an image, which gives them similarities in research methodologies. Significant progress has been achieved using classical convolution-based architectures for feature representation and learning in segmentation and object detection, such as [18, 26, 29, 32]. Recently, following the pioneering work of DETR [4], researchers have started to employ attention mechanisms to solve both tasks.

In segmentation methods using attention mechanisms, *Mask2Former* [8] stands out for using a single universal architecture to solve semantic, instance, and panoptic segmentation tasks, achieving SOTA performance and inspiring many subsequent approaches. *Mask2Former* introduces

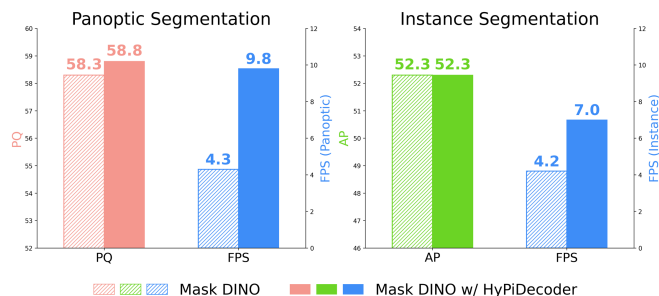


Figure 1. Performance and speed of our proposed *HyPiDecoder* incorporated *Mask DINO* with SwinL backbone on COCO. Consistent improvement can be observed in both accuracy and inference fps.

a component called the pixel decoder, which fuses and processes multi-scale features from the backbone to generate high-resolution per-pixel embeddings. As pointed out in [1], the pixel decoder accounts for over 50% of the overall computational cost in efficient networks. This component also specifically applies the multi-scale deformable attention (*MSDeformAttn*) operator proposed in [42]. Despite its relatively low FLOPs, this operator theoretically has a low compute-to-memory access ratio, and practical implementations remain suboptimal, resulting in subpar runtime performance on hardware and making the pixel decoder one of the bottlenecks of *Mask2Former*.

Based on the observations above, some recent works have identified the optimization potential in the pixel decoder, such as [1, 40], but these primarily focus on object detection. Li *et al.* [22] focuses on segmentation; however, it only reports the reduction in FLOPs of the new module without emphasizing the change in actual inference latency. Additionally, Aich *et al.* [1] can at best ensure no drop in accuracy, but cannot provide any improvement.

In this paper, inspired by a series of elegant convolution-based Feature Pyramid Network (FPN) layer designs [15, 27], we propose the *Hyper Pixel Decoder (HyPiDecoder)*. We replace some layers in the originally *MSDeformAttn*-based pixel decoder with convolution-based FPN layers, introducing explicit locality awareness into the pixel decoder.

This approach not only increases the model’s inference FPS but also enables the entire network to achieve more accurate results. In addition, we propose a multi-scale linear attention module to fuse features from different scales, and demonstrate that our pixel decoder architecture can be applied to other attention-based modules, thereby inspiring further exploration of efficient attention into the hybrid structure.

We incorporate the proposed HyPiDecoder to two strong baselines: Mask2Former, which addresses segmentation tasks, and Mask DINO, which simultaneously tackles both detection and segmentation tasks. We report results on 4 tasks: panoptic, instance, and semantic segmentation, and object detection. Our experimental results show that replacing the default pixel decoder with HyPiDecoder consistently improves inference speed, and also enhances accuracy when using comparable training epochs. Combined with Mask DINO and the SwinL backbone [28], HyPiDecoder achieves 58.8 PQ on the COCO panoptic segmentation task [25], setting a new SOTA among models using the SwinL-scale backbone without additional data, and accelerates inference fps by 127% compared to the original version of Mask DINO. On the COCO instance segmentation task, HyPiDecoder achieves 52.3 AP, matching the original Mask DINO, but with a 66% speedup.

In general, our contributions could be summarized as:

- We propose a new pixel decoder structure, HyPiDecoder, which replaces attention-based layers with convolution-based layers, resulting in reduced inference latency.
- We demonstrate the generalizability of our proposed pixel decoder by switching convolutional and attention layers. Additionally, we design a multi-scale feature processing module based on linear attention and successfully incorporate it into the HyPiDecoder.
- Our experiments confirm that the proposed HyPiDecoder can be integrated into general segmentation models and unified detection and segmentation models. The new model can achieve higher inference speeds and more accurate results.

2. Related Work

2.1. Multi-scale Feature Representation

One of the main challenges in object detection is how to effectively represent and process multi-scale features. In convolution-based methods, as a pioneering work, the Feature Pyramid Network (FPN) [26] proposes a top-down pathway to propagate information across multiple scales. Following this idea, PAFPN [27] adds an additional bottom-up pathway on top of FPN, NAS-FPN [15] stacks multi-scale feature fusion blocks and uses neural architecture search to automatically design micro-architecture of FPN. BiFPN [33] is a more easy and fast design for stacking in-

formation flow modules with neural architecture search.

In recent years, attention mechanisms is incorporated into FPN. For example, RT-DETR [40] applies attention operation to intra-scale feature maps while uses the PAFPN for inter-scale feature interactions. Deformable DETR [42] completely abandons convolutional structures and introduces a deformable attention module, which enables effective feature representation and processing without introducing too much computation. Recent work like DCNv4 [36] also explores low-level optimizations for improving the efficiency of deformable operations.

2.2. Segmentation

Segmentation is a well-studied task in computer vision. Depending on the objects being segmented, segmentation tasks can be divided into three categories: semantic segmentation, instance segmentation, and panoptic segmentation. Semantic segmentation aims to assign a class label to each pixel, while instance segmentation further distinguishes different instances of the same class. Introduced by [21], panoptic segmentation handles both thing and stuff, combining semantic segmentation and instance segmentation to provide a comprehensive understanding of the scene.

In the era of deep learning, per-pixel classification methods were once the dominant approach for semantic segmentation tasks, such as FCN [29], ASPP [6], PPM [39], and DANet [14]. Mask classification, on the other hand, is typically used in instance-level segmentation tasks, such as Mask R-CNN [18]. DETR [4] further incorporates a Transformer [34] design to handle thing and stuff segmentation simultaneously for panoptic segmentation, while MaX-DeepLab [35] introduces mask transformers for end-to-end panoptic segmentation. However, these mask classification methods often rely on bounding box predictions, are structurally complex, and can not be easily applied to semantic segmentation tasks.

Recently, MaskFormer [7] applies the mask classification paradigm to semantic segmentation, utilizing the set prediction mechanism introduced in DETR [4]. Mask2Former [8] further introduces masked attention, achieving good performance across semantic segmentation, instance segmentation, and panoptic segmentation tasks with one universal architecture.

2.3. Unified Methods

Object detection and segmentation both involve object localization, naturally sharing common model architectures and visual representations. A unified framework not only helps simplify algorithm development but also enables the joint utilization of detection and segmentation data to improve representation learning. Methods like Mask R-CNN [18], Cascade R-CNN [3], and HTC [5] typically implement this by adding a mask branch to a powerful de-

tector. Recently, Transformer-based frameworks have advanced this field. Following DETR [4], SOLQ [11] proposes a unified query representation for performing both detection and segmentation tasks. MaskDINO [23] is built upon the strong detector DINO [38] and adopts techniques such as unified and enhanced query selection, achieving optimal performance in both detection and segmentation tasks.

3. Methods

3.1. Preliminaries: MSDeformAttn

MSDeformAttn is an attention mechanism proposed in Deformable-DETR [42]. Here, we only introduce its structure in the Pixel Encoder part. MSDeformAttn is based on Deformable Attention (DeformAttn), whose formula is as follows:

$$\text{DeformAttn}(z_q, p_q, x) = \sum_{m=1}^M W_m \left[\sum_{k=1}^K A_{mqk} \cdot W'_m x(p_q + \Delta p_{mqk}) \right] \quad (1)$$

where, $x \in \mathbb{R}^{C \times H \times W}$ is the feature map, M is the number of heads, W_m is the sampling weight for each head, K is the number of sampled points per head, p_q is the coordinate position of the recurrent current reference point, Δp_{mqk} and A_{mqk} are the predicted sampling offsets and normalized attention weights, respectively, both obtained through a linear transformation of z_q . $x(p_q + \Delta p_{mqk})$ is the interpolated result at the corresponding position, W'_m is a projection matrix. This formula can be interpreted as predicting multiple offsets and normalized attention weights for each point, then performing an attention operation among the offset positions according to the attention weights.

MSDeformAttn extends DeformAttn by introducing multiple scales. The formula is as follows:

$$\text{MSDeformAttn}(z_q, \hat{p}_q, \{x^l\}_{l=1}^L) = \sum_{m=1}^M W_m \left[\sum_{l=1}^L \sum_{k=1}^K A_{mlqk} \cdot W'_m x^l(\phi_l(\hat{p}_q) + \Delta p_{mlqk}) \right] \quad (2)$$

where L is the number of scales, $\hat{p}_q$ is the reference point after projecting the current point to different scales. MSDeformAttn differs from DeformAttn in that its input is not a single feature map x , but multiple feature maps $\{x^l\}_{l=1}^L$, and the sampling process, attention weight prediction, normalization of the attention weights etc., are all performed across multiple scales.

The MSDeformAttn Pixel Decoder structure, proposed in [42] and utilized in works such as [8, 19, 23, 38], simply stacking multiple MSDeformAttn blocks together. The input consists of multi-scale feature maps from the back-

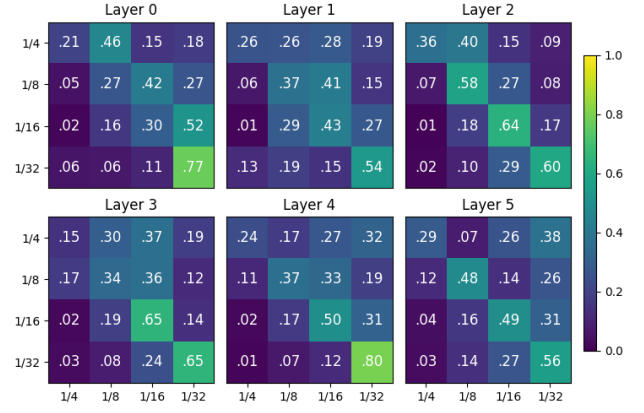


Figure 2. Deformable Attention Weight Distribution from ResNet-50 Mask DINO Pixel Decoder. Each row represents the distribution of attention weights from the corresponding scale’s output.

bone, and the output is a multi-scale feature map that has completed multi-scale information interaction.

In MSDeformAttn, each output pixel is obtained by interpolating from the input feature map with non-fixed offsets, resulting in a low computation-to-memory-access ratio, making it difficult to implement efficiently. Although MSDeformAttn is a low-FLOPS operator in theory, it has a actual high runtime latency. Therefore, we need more mature and efficient operators to replace MSDeformAttn.

3.2. Locality Awareness

For the classic attention module [34], a notable characteristic is its permutation invariance when no position embedding is added, as each token is treated equally. In the context of visual tasks, this means it lacks locality awareness—nearby pixels and distant pixels are treated equally, which is unfavorable for visual tasks. In visual tasks, closer pixels in space should have stronger correlation. Introducing this bias can significantly optimize model design and improve performance. This intuition aligns with human perception, guiding the parameter-sharing design of CNNs compared to MLPs, which have been proven effective in computer vision over the past decades.

Similarly in MSDeformAttn, although it employs both a learnable positional encoding to represent the scale index and a 2D sinusoidal positional encoding to represent coordinates, it still fails to fully exploit the locality bias. Intuitively, we believe that for MSDeformAttn, the output at certain level should obtain the maximum amount of information from the same level of the input feature map, i.e. the sum of the attention weights should be maximized. For Mask DINO with a ResNet-50 backbone, we calculate and aggregate the attention weights from different layers at each scale to other scales. Our experiment are conducted on COCO val2017, and the results are shown in Figure

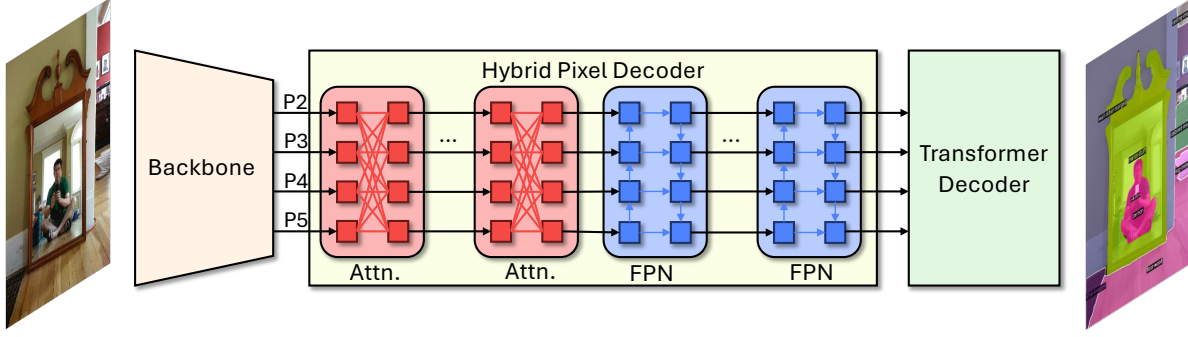


Figure 3. The architecture of Hybrid Pixel Decoder (HyPiDecoder).

2, where each row represents the source distribution of attention weights for the corresponding scale’s output. We observe that although the weights on the diagonal are relatively large, their values are not necessarily the largest; this is especially true for the 1/4 scale, which tends to allocate more weights to higher scales. Therefore, in MSDeformAttn, features from lower scales do not focus on local information as we anticipate but instead allocate a substantial amount of weight to more global information. It is noteworthy that since MSDeformAttn flattens and concatenates pixels from different scales, the lowest 1/4 scale pixels account for over 75% of the total pixel count, meaning a large number of pixels are influenced by this global information, while the highest 1/32 scale pixels only account for about 1% of the total pixel count. We can foresee that this use of information distribution is relatively inefficient.

3.3. Hybrid Pixel Decoder (HyPiDecoder)

To address the issues mentioned earlier, including the lack of efficient hardware implementation for MSDeformAttn and its deficiency in locality-awareness, we propose the Hybrid Pixel Decoder (HyPiDecoder). The architecture overview of HyPiDecoder is shown in Figure 3. Similar to MSDeformAttn Pixel Decoder, we use a multi-layered architecture, which can be seamlessly replace the MSDeformAttn Pixel Decoder. The main difference is that: our proposed HyPiDecoder replaces part of the MSDeformAttn blocks with convolution-based FPN blocks. Additionally, our architecture is customizable; we can implement our design with suitable attention blocks or convolution blocks as long as the input and the output are feature maps in multiple scales.

3.3.1. Convolution-based FPN Blocks

Since both the pixel decoder and the feature pyramid network (FPN) component in convolution-based detection and segmentation models handle the representation and learning of multi-scale feature maps, we can naturally borrow some model design techniques from FPN.

Specifically, in this work, we borrow the Cross Stage

Partial (CSP) connections proposed in YOLOv4 [2] and, inspired by the structure of RT-DETR [40], replace the 3×3 convolution layers in the CSP module with the RepVGG [10] module. The RepVGG module splits a 3×3 convolution into a 3×3 convolution and a 1×1 convolution during training and then reverts it to a 3×3 convolution during inference, enhancing model accuracy without affecting inference speed. We refer to the modified module as CSPRepLayer.

The structure of this CSPRepLayer is illustrated in Figure 4. In the top-down pathway, we perform a stride-2 convolution downsampling on higher-level features, concatenate them with lower-level features, and pass them through the CSPRepVGG module. In the bottom-up pathway, we apply bilinear interpolation to upsample the lower-scale feature map, concatenate it with features at the current scale, and then pass it through a CSP module to obtain the final output. This structure can also be regarded as a variant of PAFPN [27].

Compared to the MSDeformAttn module, which uses shared attention weight and offset mapping layer across each level, CSPRepVGG module significantly increases the parameter count due to the introduction of level-independent standard 3×3 convolutions, we can use separable convolutions (a cascade of 1×1 convolutions and depthwise 3×3 convolutions) to replace the standard 3×3 convolutions if parameter count is a critical factor. We refer to this module as the CSP Layer with Separable Convolution (CSPSepLayer).

3.3.2. Multi-scale Linear Attention

To demonstrate the extensibility of our proposed HyPiDecoder module, we aim for it to function effectively even without MSDeformAttn. However, due to the lack of an off-the-shelf multi-scale attention module other than MSDeformAttn, we attempted to propose a new multi-scale attention module called Multi-scale Linear Attention (MSLinearAttn). Next, we will introduce this module.

In the pioneering work of [34], the output of the proposed attention mechanism can be defined by the following

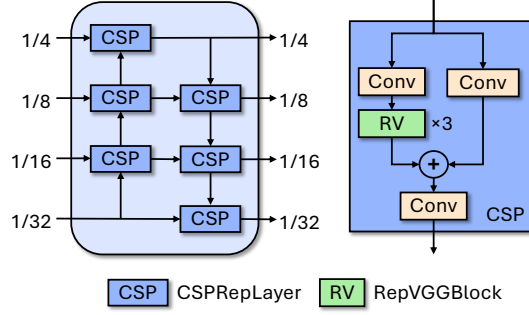


Figure 4. The structure of Cross Stage Partial RepVGG Layer (CSPRep).

equation:

$$O_i = \sum_{j=1}^N \text{softmax}(Q_i, K_j) V_j \quad (3)$$

where N is the number of tokens and D is the embedding dimension. $Q_i, K_j, V_j \in \mathbb{R}^D$ is one row from the query, key and value matrix respectively, $O_i \in \mathbb{R}^D$ is the i -th row returned as a vector. The output of attention mechanism is a weighted average of value vectors, where the weight is normalized to sum to 1 as a property of softmax. We can rewrite Eq. 3 as follows:

$$O_i = \frac{\sum_{j=1}^N \text{sim}(Q_i, K_j) V_j}{\sum_{j=1}^N \text{sim}(Q_i, K_j)} \quad (4)$$

where sim is a function representing similarity between the query embedding and key embedding. Linear Attention is initially proposed by [20] following Eq. 4 where sim is defined as:

$$\text{sim}(Q_i, K_j) = \phi(Q_i)^T \phi(K_j) \quad (5)$$

where ϕ can be any element-wise non-linear function. Further simplification can be applied to Eq. 4 as follows:

$$O_i = \frac{\phi(Q_i)^T \sum_{j=1}^N \phi(K_j) V_j^T}{\phi(Q_i)^T \sum_{j=1}^N \phi(K_j)} \quad (6)$$

The classic softmax attention defined by Eq. 3 has a time complexity of $O(N^2D)$, while linear attention calculated using Eq. 6 has a time complexity of $O(ND^2)$, which is far less than softmax attention when $N \gg D$. This condition is a common scenario in vision tasks like object detection and segmentation, where the number of tokens is the number of pixels in the feature map, which can be in the order of 10^4 , while the embedding dimension is usually 256 or 512. Therefore, linear attention is more computationally affordable than softmax attention in our application.

Based on the definitions above, we propose the Multi-scale Linear Attention (MSLinearAttn) block, as shown in

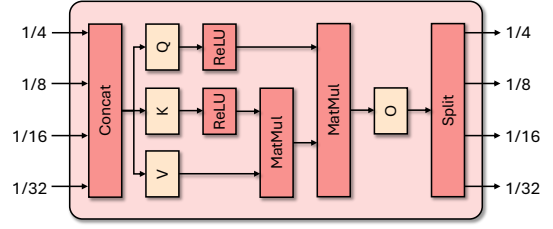


Figure 5. The structure of Multi-scale Linear Attention Layer (MSLinearAttn).

Figure 5. The input to this module is $\{x^l\}_{l=1}^L$, where L is the number of scales, and x_i has the shape (D, H_i, W_i) . Note that the feature maps from different scales have the same number of channels, D . First, we flatten each scale's feature map and concatenate them along the channel dimension, resulting in a combined tensor x with the shape $(D, \sum_{i=1}^L H_i W_i)$. We then pass this x through a linear layer to obtain the query, key, and value, which are then fed into Eq. 6, yielding an output tensor. This output tensor is further processed through an output linear layer to produce y . Finally, we split y into L different scales, with each feature map having the same shape as corresponding input.

Similarly to softmax attention, linear attention is also permutation invariant. Therefore, we need to introduce positional embeddings to incorporate positional information into the input. Specifically, we use the following form of positional embedding:

$$\hat{e}_{l,x,y} = \bar{e}_l + e_{x,y} \quad (7)$$

where $e_{x,y}$ is a 2D sinusoidal position embedding at each scale, and $\bar{e}_l$ is a scale-dependent learnable embedding. We add the aforementioned positional embedding to the input feature map before being fed into the linear attention. We employ a block-dependent $\bar{e}_l$ in our implementation.

4. Experiments

We integrate the proposed HyPiDecoder into both the Mask2Former [8] and Mask DINO [23] frameworks, comparing the performance of models with HyPiDecoder against the original frameworks. Extensive experiments are conducted across four popular tasks: object detection, instance segmentation, and panoptic segmentation on the COCO [25] dataset, and semantic segmentation on the ADE20K [41] and Cityscapes [9] datasets. For our experiments, we use ResNet-50 [17] and SwinL [28] as backbones. All experiments are conducted with a batch size of 16 on A100 GPUs. We utilize 4 GPUs for ResNet-50 and 8 GPUs for SwinL.

We used multiple HyPiDecoder layer configurations in our experiments, indicated by subscripts to represent the specific setup: $nAmB$ denotes using n A blocks followed

Model	Epochs	#Query	PQ	PQ Th	PQ St	Box AP Th _{pan}	Mask AP Th _{pan}	mIoU	GFLOPS	Params	FPS
ResNet-50 backbone											
DETR [4]	500+25	100	43.4	48.2	36	-	31.1	-	-	-	-
Panoptic Segformer [24]	24	353	49.6	54.4	42.4	-	41.7	-	-	-	-
kMaX-DeepLab [37]	81	128	53.0	58.3	44.9	-	-	-	-	-	-
MaskConver [31]	300	100	53.6	58.9	45.6	-	-	-	-	-	-
Mask2Former [8]	50	100	51.9	57.7	43.0	-	41.7	-	226	44M	20.5
Mask2Former [8] w/ default PD	12	100	47.0	52.3	38.9	-	37.4	58.3	226	44M	20.5
Mask2Former w/ HPD _{3d3s} (ours)	12	100	47.5	53.4	38.7	-	37.8	58.9	326	54M	27.2 (+32%)
Mask DINO [23] w/ default PD	50	300	53.0	59.1	43.9	48.8	44.3	60.7	283	52M	17.1
Mask DINO w/ HPD _{3l3r} (ours)	50	300	53.4	59.2	44.6	49.2	44.8	61.4	482	91M	20.5 (+20%)
Mask DINO w/ HPD _{3d3r} (ours)	50	300	53.7	60.0	44.2	49.3	44.9	61.8	468	91M	21.8 (+27%)
SwinL backbone											
OneFormer [19]	100	150	57.9	64.4	48.0	-	49.0	-	-	-	-
Mask2Former [8]	100	200	57.8	64.2	48.1	-	48.6	67.4	868	216M	7.8
Mask2Former [8] w/ default PD	50	200	57.6	63.9	48.2	-	48.2	67.8	868	216M	7.8
Mask2Former w/ HPD _{3d3s} (ours)	50	200	58.1	64.5	48.4	-	48.6	67.7	941	225M	9.6 (+23%)
Mask DINO [23] w/ default PD *	50	300	58.3	65.1	48.0	56.2	50.6	67.5	1355	223M	4.3
Mask DINO w/ HPD _{3l3r} (ours) *	50	300	58.6	65.1	48.9	56.2	50.5	68.0	1215	278M	8.8 (+104%)
Mask DINO w/ HPD _{3d3r} (ours) *	50	300	58.8	65.2	49.1	56.6	50.9	67.8	1199	278M	9.8 (+127%)

Table 1. Results for HyPiDecoder incorporated Mask2Former [8] and Mask DINO [23] and other models with ResNet-50 and SwinL backbone on **panoptic segmentation on COCO val2017**. PD and HPD are abbreviations for Pixel Decoder and HyPiDecoder, respectively. * Different from Mask DINO [23] which use 5 scales for the run with a SwinL backbone, we only use 4 scales for this SwinL run.

by m B blocks consecutively. n and m are integer numbers and A , B can be either one of l , d , r and s , where l represents MSLinearAttn, d represents MSDeformAttn, r represents CSPLayer with a RepVGG (CSPRep) block, and s represents CSPLayer with a separable convolution (CSPSep) block.

Unless otherwise specified, all our profiling results are obtained on a single Nvidia A100. We serialize the models to ONNX format, and report FPS with an input size of (800, 1120), which is the average image size from COCO val2017 after preprocessing. If there are RepVGG blocks in the model, we replace each of them with a single equivalent convolution layer during the serialization.

4.1. Main Results

4.1.1. Panoptic segmentation

The results of panoptic segmentation on COCO val2017 are shown in Table 1. For SwinL Mask DINO adapted with HyPiDecoder_{3d3r}, we achieved a PQ of 58.8 (+0.5) with a 127% improvement in inference speed. This establishes a new SOTA performance on the SwinL-scale backbone without additional training data. In experiments with SwinL Mask2Former, we obtained a PQ score of 58.1 (+0.5) with a 23% improvement in inference speed. Notably, our model trained with 50 epochs achieved slightly better PQ than the Mask2Former baseline trained for 100 epochs. Similar improvements can be observed on the ResNet-50 backbone. Our proposed HyPiDecoder consistently outperforms the default pixel decoders used in Mask2Former and Mask DINO across PQ, AP, and mIoU, with faster inference speed, underscoring the efficiency of HyPiDecoder. An interesting finding is that two ex-

periments using $3l3r$ demonstrate that HyPiDecoder can achieve both higher speed and accuracy without utilizing MSDeformAttn.

Since HyPiDecoder replaces the MSDeformAttn module with convolution-based FPN layers, and these FPN layers have more FLOPS and number of parameters. We observe an increase in FPS but also an increase in both FLOPS and number of parameters. FLOPS is an indirect metric for describing inference speed; when reported alongside FPS, more emphasis should be placed on FPS. In scenarios where strict constraints on parameter count are required, the separable convolution version of HyPiDecoder should be used, as it introduces fewer additional parameters compared to the version using RepVGG blocks.

4.1.2. Instance segmentation and object detection

The results of instance segmentation and object detection are shown in Table 2. Since Mask DINO is already a very strong model for instance segmentation and object detection, introducing our HyPiDecoder achieve Mask AP of 52.3 and no further improvement on instance segmentation or object detection. However, we observe a 66% improvement in FPS compared to the origin version of Mask DINO.

On the relatively weak SwinL Mask2Former model with 50 training epochs, we observe that introducing HyPiDecoder achieves an AP of 49.9 (+0.4) and provides improvements in both AP₅₀ and AP₇₅. We also observe similar improvement on the ResNet-50 backbone under the 12-epoch training schedule.

4.1.3. Semantic segmentation

We present the semantic segmentation results on the ADE20K [41] in Table 3 and Cityscapes [9] in Table 4, re-

Model	Epochs	#Query	Mask AP	Box AP	AP ₅₀ ^{mask}	AP ₇₅ ^{mask}	AP _S ^{mask}	AP _M ^{mask}	AP _L ^{mask}	GFLOPS	Params	FPS
ResNet-50 backbone												
Mask-RCNN [12, 16, 18]	400	Dense	42.5	48.2	-	-	23.8	45.0	60.0	207	40M	10.3 [‡]
HTC [5]	36	Dense	39.7	44.9	61.4	43.1	22.6	42.2	50.6	441	80M	5.0 [‡]
QueryInst [13]	36	300	40.6	45.6	63.0	44.0	23.4	42.5	52.8	-	-	7.0 [‡]
DINO (4scale) [38]	36	900	-	50.9	-	-	-	-	-	245	47M	19.6 [‡]
Mask2Former [8]	50	100	43.7	46.2 [†]	66.0	46.9	23.4	47.2	64.8	226	44M	20.5 [‡]
DI-MaskDINO [30]	50	300	46.7	51.9	-	-	27.5	49.8	66.2	-	-	-
Mask DINO [23]	50	300	46.3	51.7	69.0	50.7	26.1	49.3	66.1	286	52M	14.2
Mask2Former [8] w/ default PD	12	100	39.1	-	60.3	42.1	17.9	42.0	60.4	226	44M	20.5
Mask2Former w/ HPD _{3d3s} (ours)	12	100	39.4	-	60.3	41.6	20.0	42.2	60.6	326	54M	27.2 (+33%)
SwinL backbone												
HTC++ [5, 28]	72	Dense	49.5	57.1	-	-	31.0	52.4	67.2	1470	284M	-
DINO [38]	36	900	-	58.5	-	-	-	-	-	1285	217M	8.1 [‡]
Mask2Former [8]	100	200	50.1	-	-	-	29.9	53.9	72.1	868	216M	7.8
DI-MaskDINO [30]	50	300	51.8	57.8	-	-	31.8	55.1	72.2	-	-	-
Mask2Former [8] w/ default PD	50	200	49.5	-	73.8	53.6	29.2	53.2	72.1	868	216M	7.8
Mask2Former w/ HPD _{3d3s} (ours)	50	200	49.9	-	74.1	54.2	29.0	53.7	72.0	941	225M	9.6 (+23%)
Mask DINO [23] w/ default PD *	50	300	52.3	59.0	76.6	57.8	33.1	55.4	72.6	1326	223M	4.2 [◊]
Mask DINO w/ HPD _{3d3r} (ours) *	50	300	52.3	58.7	76.6	57.2	32.2	55.4	72.3	1761	281M	7.0 (+66%)

Table 2. Results for HyPiDecoder incorporated Mask2Former [8] and Mask DINO [23] and other models with ResNet-50 and SwinL backbone on **object detection and instance segmentation on COCO val2017** without extra data or tricks. PD and HPD are abbreviations for Pixel Decoder and HyPiDecoder, respectively.

* Following Mask DINO [23], we use 5 scales for large Mask DINO models with a SwinL backbone, and use 4 scales for other models.

[‡] A FPS with this label indicates that it is copied from [23].

[◊] Li *et al.* [23] reported FPS of 5.6 is not reproducible while the FPS measured using their method is 2.6, which is lower than the value in the table, and thus not used.

method	backbone	mIoU (s.s.)	mIoU (m.s.)
Mask2Former [8] w/ default PD	R50	47.2	49.2
Mask2Former w/ HPD _{3d3s} (ours)	R50	47.6	49.7

Table 3. **Semantic segmentation on ADE20K val** with 150 categories.

method	backbone	mIoU (s.s.)	mIoU (m.s.)
Mask2Former [8] w/ default PD	R50	79.4	82.2
Mask2Former w/ HPD _{3d3s} (ours)	R50	81.1	82.7

Table 4. **Semantic segmentation on Cityscapes val**.

spectively. We use Mask2Former with a ResNet-50 backbone as the baseline. Under the 160k training iterations on ADE20K, HyPiDecoder achieves an mIoU of 47.6 (+0.4) in single-scale and 49.7 (+0.5) in multi-scale testing. Under the 90k training iterations on CityScapes, HyPiDecoder obtains an mIoU of 81.1 (+0.7) in single-scale and 82.7 (+0.5) in multi-scale testing, showing consistent improvements compared to the original version of Mask2Former.

4.2. Ablation Studies

Number of Layers. Table 5 ablates number of layers with HyPiDecoder incorporated ResNet-50 MaskDINO on COCO val2017 panoptic segmentation in 50 epochs training schedule. We observe a modest decrease in model accuracy with a reduction in the number of layers, accompanied by a steady improvement in inference FPS.

Attention Module. Table 6 ablates the attention module in hybrid layers in HyPiDecoder incorporated ResNet-50 MaskDINO on COCO val2017 panoptic segmentation in 12 epochs training schedule. We can see that the differences between the MSLinearAttn and MSDeformAttn modules are minimal, and MSLinearAttn is even slightly better in PQ. Interestingly, this conclusion is not entirely consistent with that in Table 1, which may be due to differences in the backbone or training epochs.

Attention Layer Ratio. Table 7 ablates the number of attention layers with a limitation of stacking 6 blocks in HyPiDecoder incorporated ResNet-50 MaskDINO on COCO val2017 panoptic segmentation in 12 epochs training schedule. We observe that using more or fewer attention layers are both worth considering, but using only one single type of layer, especially MSLinearAttn alone, results in a significant drop in performance.

Layer Order inside Hybrid Blocks. Table 8 ablates the order of hybrid layers in HyPiDecoder incorporated ResNet-50 MaskDINO on COCO val2017 panoptic segmentation in 12 epochs training schedule. We observe that the default 3/3r configuration performs optimally, while placing the attention block after the FPN block, or alternating the two, tends to harm performance.

Mixed Attention Module. Table 9 ablates the mixed attention module in HyPiDecoder incorporated ResNet-50 MaskDINO on COCO val2017 panoptic segmentation in 12 epochs training schedule. We observe that using a mix

Layers	PQ	AP	mIoU	FPS
1l1r	50.5	41.8	58.7	36.7
2l2r	52.4	44.1	60.3	27.7
3l3r	53.4	44.8	61.4	20.5

Table 5. Ablation on number of layers with HyPiDecoder incorporated MaskDINO on COCO val2017 panoptic segmentation.

Layers	PQ	AP	mIoU
6l0r	43.7	34.0	53.5
2l4r	49.2	41.0	56.9
3l3r	49.3	40.9	57.6
4l2r	48.9	40.7	56.8
0l6r	48.7	41.0	56.5

Table 7. Ablation on ratio of attention module with HyPiDecoder incorporated MaskDINO on COCO val2017 panoptic segmentation.

Layers	PQ	AP	mIoU
3l3r	49.3	40.9	57.6
3r3l	48.6	40.3	57.0
lrlrlr	48.9	40.7	57.4

Table 8. Ablation on layer order with HyPiDecoder incorporated MaskDINO on COCO val2017 panoptic segmentation.

Layers	PQ	AP	mIoU
3d3r	49.0	41.0	57.1
3l3r	49.3	40.9	57.6

Table 6. Ablation on attention module with HyPiDecoder incorporated MaskDINO on COCO val2017 panoptic segmentation.

Layers	PQ	AP	mIoU
4l2r	48.9	40.7	56.8
2d2l2r	48.7	40.1	56.9
2l2d2r	48.7	40.2	56.9

Table 9. Ablation on mixed attention module with HyPiDecoder incorporated MaskDINO on COCO val2017 panoptic segmentation.

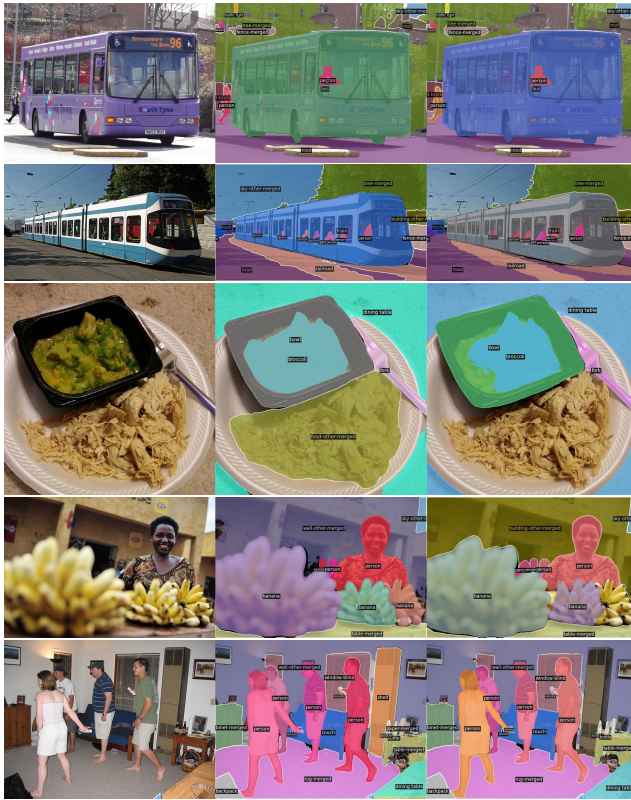


Figure 6. SwinL panoptic segmentation results on COCO val2017: **left**: origin image, **middle**: Mask DINO with HyPiDecoder, **right**: Mask DINO with default pixel decoder. Best viewed in color and zoomed in.

of the two different attention modules can also yield meaningful results, but it does not appear to be optimal.

Qualitative analysis We provide qualitative results in Figure 6 on COCO val2017 panoptic segmentation on SwinL backbone. We observe that the model with HyPiDecoder produces more accurate and detailed segmentation results compared to the model with the default pixel decoder.

5. Conclusion

In this paper, we propose HyPiDecoder, which introduces explicit locality information by replacing part of the MSDeformAttn layers with convolution-based FPN layers, significantly boosting the inference speed and accuracy of segmentation models. HyPiDecoder can seamlessly integrate with current popular segmentation models, enabling it to perform object detection, semantic segmentation, instance segmentation, and panoptic segmentation tasks. Our experimental results show that, integrated to Mask DINO with SwinL backbone, HyPiDecoder achieves 58.8 PQ on the COCO panoptic segmentation task, setting a new SOTA for models using SwinL-scale backbone without additional data, and improves inference speed by 127% compared to the original version.

Limitations: In this work, we do not extend our proposed HyPiDecoder to pure detection models, such as DINO [38], or to universal image segmentation models, such as OneFormer [19]. Additionally, we do not apply our proposed model to larger pretraining datasets, larger backbones or larger training epochs. We leave this to our future work.

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